Github REPO: https://github.com/MChaudhary123/Sprint-1---Foundations-of-SWE

1. User Stories (3 points)

ID	User Story Name	User Story Description	Priority	Effort (hrs)
1	Choose a board size	As a player, I want to choose the board size before starting a game so that I can play on a smaller or larger board.	High	3
2	Choose the board type	As a player, I want to choose the board type (English, Hexagon, or Diamond) so that I can play the version of Solitaire I prefer.	High	4
3	Start a new game of the chosen board size and type	As a player, I want to start a new game with the board size and type I chose so that I begin with a full board and an empty center hole.	High	5
4	Make a move in a game	As a player, I want to jump a peg over an adjacent peg into an empty hole so that the jumped peg is removed and I get closer to leaving one peg.	High	8
5	A game is over	As a player, I want the game to tell me when no more valid moves are possible and show my rating so that I know the game is over and how well I did.	Medium	4

2. Acceptance Criteria (AC) (12 points)

User Story	AC ID	Description of Acceptance Criterion	Status
1. Choose a board size	1.1	**AC 1.1 Choose a valid board size**	toDo

User Story	AC ID	Description of Acceptance Criterion	Status
		Given the player is on the start screen and no game is in progress When the player enters a board size of 7 (an odd number from 5 to 9) Then the board size is set to 7 and the board is drawn at that size	
	1.2	**AC 1.2 Choose an invalid board size** Given the player is on the start screen When the player enters a board size that is even, below 5, above 9, or not a number Then an error message is shown and the board size stays at its previous value	toDo
	1.3	**AC 1.3 Default board size** Given the program has just been opened When the player has not entered a board size Then the board size is 7	toDo
2. Choose the board type	2.1	**AC 2.1 Choose the English board** Given the player is on the start screen When the player selects the "English" radio button Then the board is drawn in the English (cross) shape	toDo
	2.2	**AC 2.2 Choose the Hexagon board** Given the player is on the start screen When the player selects the "Hexagon" radio button Then the board is drawn in the Hexagon shape	toDo
	2.3	**AC 2.3 Choose the Diamond board** Given the player is on the start screen When the player selects the "Diamond" radio button Then the board is drawn in the Diamond shape	toDo
	2.4	**AC 2.4 Default board type** Given the program has just been opened When the player has not selected a board type Then the English board type is selected	toDo

User Story	AC ID	Description of Acceptance Criterion	Status
3. Start a new game of the chosen board size and type	3.1	**AC 3.1 Start a new game** Given the player has chosen a valid board size and a board type When the player clicks "New Game" Then every hole on the board is filled with a peg except the center hole, which is empty	toDo
	3.2	**AC 3.2 Start a new game while a game is in progress** Given a game is in progress and some pegs have been removed When the player clicks "New Game" Then the old game ends and a fresh board is set up with the currently chosen size and type	toDo
4. Make a move in a game	4.1	**AC 4.1 Make a valid orthogonal move** Given a game is in progress, a peg is in the selected hole, the adjacent hole (up, down, left, or right) has a peg, and the hole two positions away is empty When the player selects the peg and then selects the empty hole two positions away Then the peg moves to the empty hole and the jumped peg is removed from the board	toDo
	4.2	**AC 4.2 Make a valid diagonal move** Given a game is in progress, a peg is in the selected hole, the diagonally adjacent hole has a peg, and the hole two positions away on that diagonal is empty When the player selects the peg and then selects that empty hole Then the peg moves to the empty hole and the jumped peg is removed from the board	toDo
	4.3	**AC 4.3 Make an invalid move** Given a game is in progress When the player tries to move a peg to a hole that is not empty, not two positions away, or has no peg in between	toDo

User Story	AC ID	Description of Acceptance Criterion	Status
		Then the move is rejected and the board does not change	
5. A game is over	5.1	**AC 5.1 Game over with one peg left** Given a game is in progress When the player makes a move that leaves only one peg on the board Then the game ends and the message "Game over - Outstanding" is shown	toDo
	5.2	**AC 5.2 Game over with no valid moves left** Given a game is in progress and more than one peg is on the board When the player makes a move after which no valid move is possible Then the game ends and a rating is shown based on the pegs left: Very Good (2), Good (3), or Average (4 or more)	toDo
	5.3	**AC 5.3 Game continues while moves remain** Given a game is in progress When the player makes a move and at least one valid move is still possible Then the game does not end and the player can keep playing	toDo

III. LLM Usage

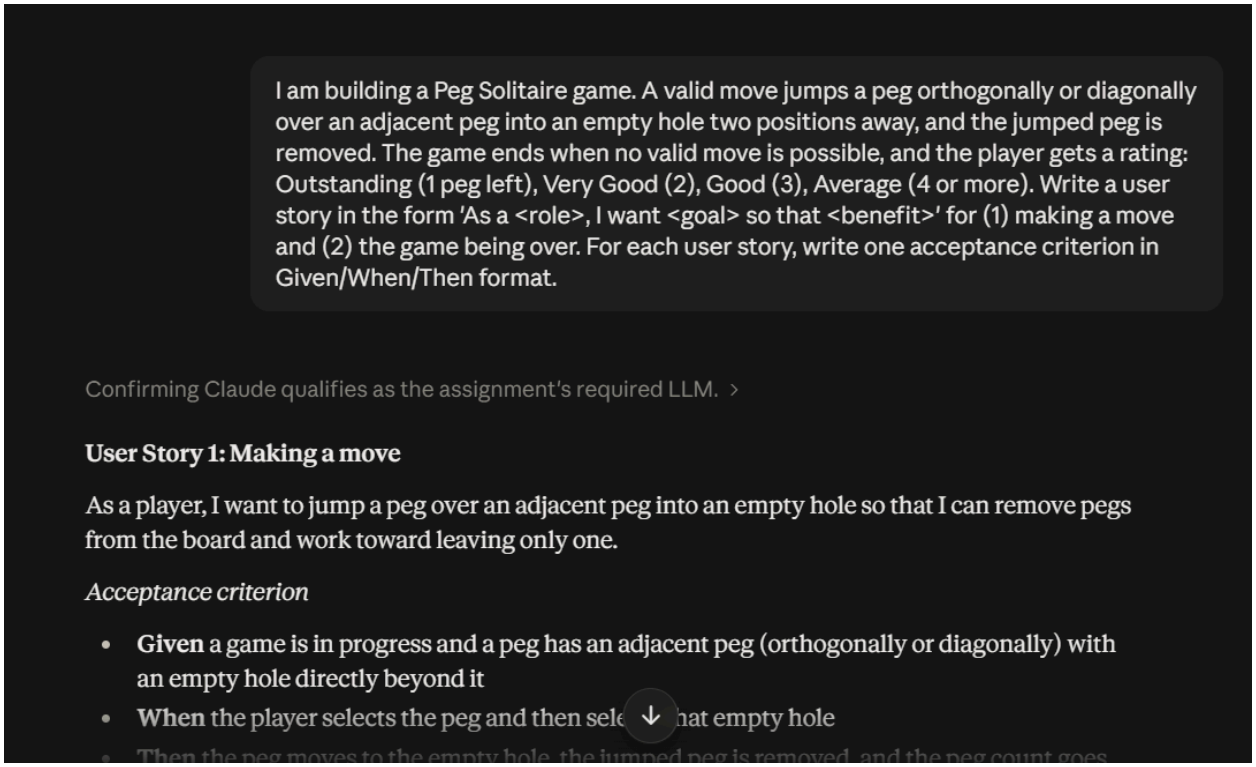
LLM used: Claude. The LLM was used for User Story 4 (Make a move in a game) and User Story 5 (A game is over), and for one acceptance criterion for each. The versions in the tables above are the corrected versions.

Prompt used:

"I am building a Peg Solitaire game. A valid move jumps a peg orthogonally or diagonally over an adjacent peg into an empty hole two positions away, and the jumped peg is removed. The game ends when no valid move is possible, and the player gets a rating: Outstanding (1 peg left),

Very Good (2), Good (3), Average (4 or more). Write a user story in the form 'As a <role>, I want <goal> so that <benefit>' for (1) making a move and (2) the game being over. For each user story, write one acceptance criterion in Given/When/Then format."

Screenshots of the LLM prompt and answer:



User Story 1: Making a move

As a player, I want to jump a peg over an adjacent peg into an empty hole so that I can remove pegs from the board and work toward leaving only one.

Acceptance criterion

- **Given** a game is in progress and a peg has an adjacent peg (orthogonally or diagonally) with an empty hole directly beyond it
- **When** the player selects the peg and then selects that empty hole
- **Then** the peg moves to the empty hole, the jumped peg is removed, and the peg count goes down by one

User Story 2: Game over

As a player, I want the game to detect when no valid moves remain and show my rating so that I know the game has ended and how well I played.

Acceptance criterion

- **Given** a game is in progress
- **When** the player makes a move that leaves no valid moves on the board
- **Then** the game ends and displays a rating based on the pegs remaining: Outstanding (1), Very Good (2), Good (3), or Average (4 or mor

Errors made by the LLM and how they were corrected:

The acceptance criterion for making a move was too broad. It combined orthogonal and diagonal jumps into one scenario, which makes it hard to test. I split it into AC 4.1 (valid orthogonal move) and AC 4.2 (valid diagonal move).

The LLM only covered a valid move and did not say what happens when the player tries an illegal one. I added AC 4.3, where the move is rejected and the board does not change.

The acceptance criterion for game over mixed two different cases: finishing with one peg left, and getting stuck with several pegs left. I separated them into AC 5.1 (one peg left, rated Outstanding) and AC 5.2 (no valid moves left, rated by pegs remaining).

The LLM did not cover the case where the game should keep going. I added AC 5.3, where the game continues while at least one valid move remains.